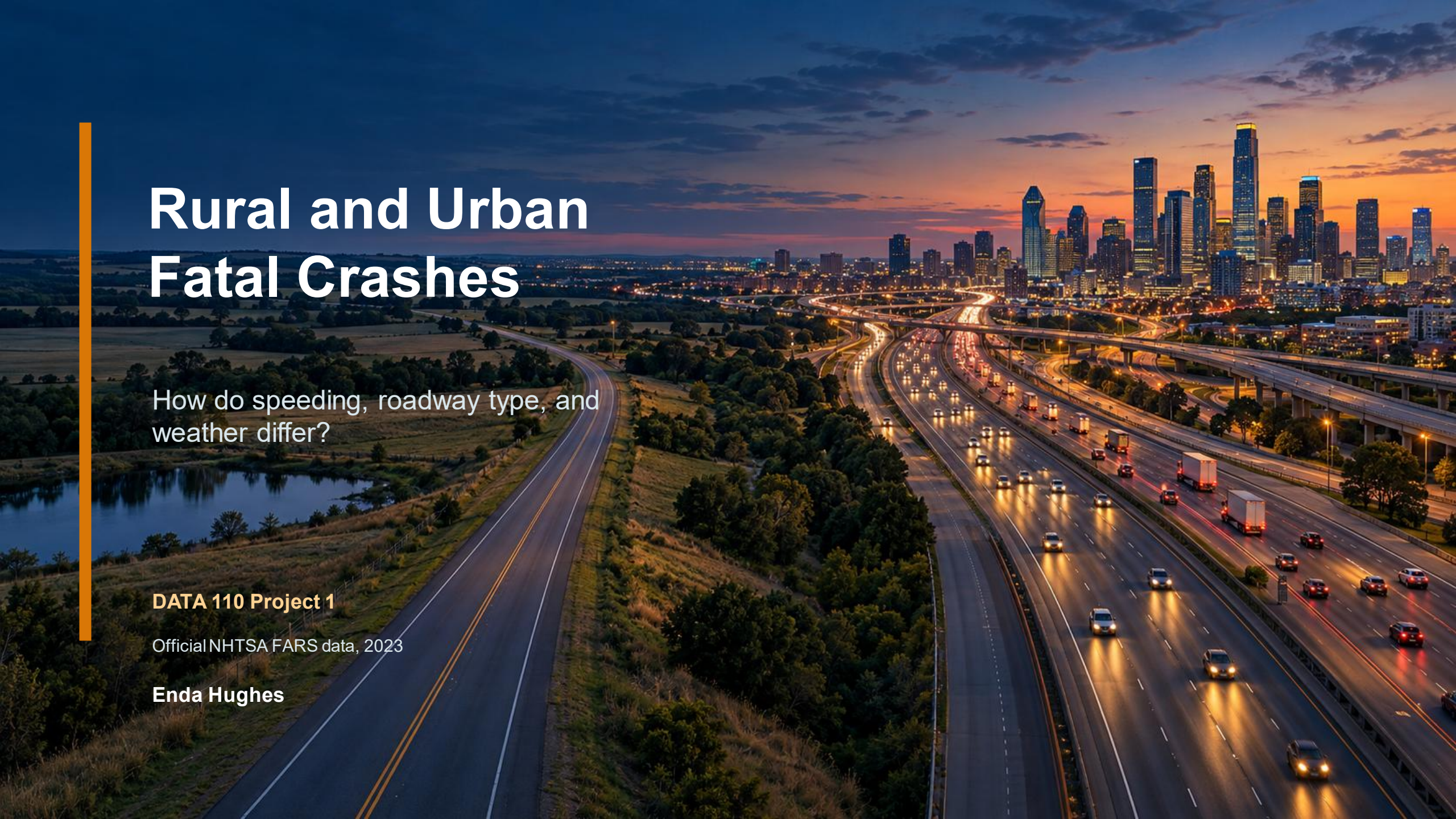


Rural and Urban Fatal Crashes

How do speeding, roadway type, and weather differ?

DATA 110 Project 1

Official NHTSA FARS data, 2023

Enda Hughes

Research question and official data

How do fatal crashes on rural roads differ from those on urban roads?

The comparison focuses on speeding involvement, roadway type, weather, and fatalities per crash.

37,769

fatal crashes in the national file

50 states + D.C.

FARS is a census of qualifying fatal traffic crashes

1 crash per row

Vehicle records supply speeding involvement

FARS compiles information from police reports, roadway records, death certificates, toxicology reports, and related state records.

Structure, missingness, and cleaning

37,769
raw crash records

37,580
classified as rural or urban

189
excluded for unknown location

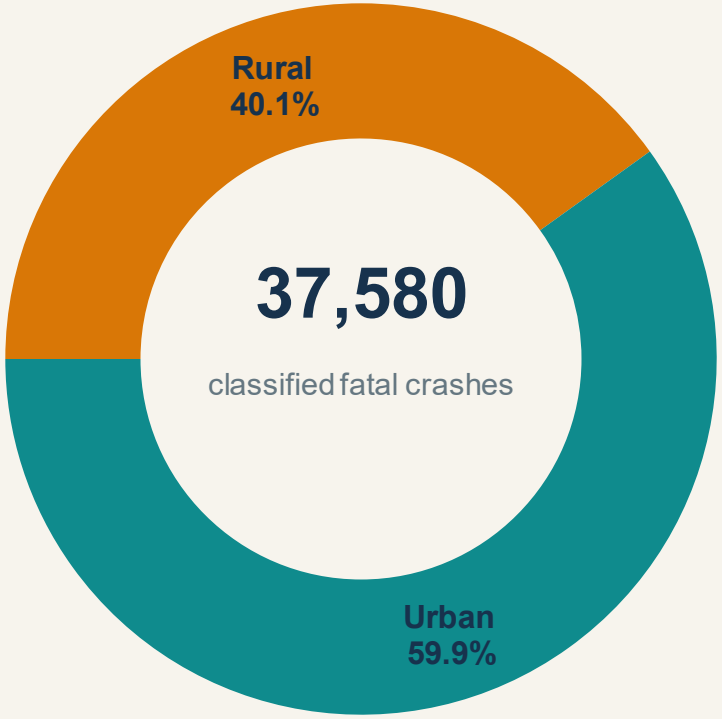
3,022
unknown speeding status

Cleaning decisions

- Keep only crashes coded Rural or Urban
- Mark a crash speed-related when any involved vehicle has a positive speeding code
- Exclude unknown speeding status from the speeding percentage
- Combine functional classes into freeway, arterial, collector, and local roads
- Group rain, snow, fog, wind, sleet, and freezing precipitation as adverse weather

Unit of analysis: one fatal crash. FARS does not contain total miles driven, so the charts compare recorded crashes rather than risk per mile.

Urban roads contain most recorded fatal crashes



22,516

urban fatal crashes

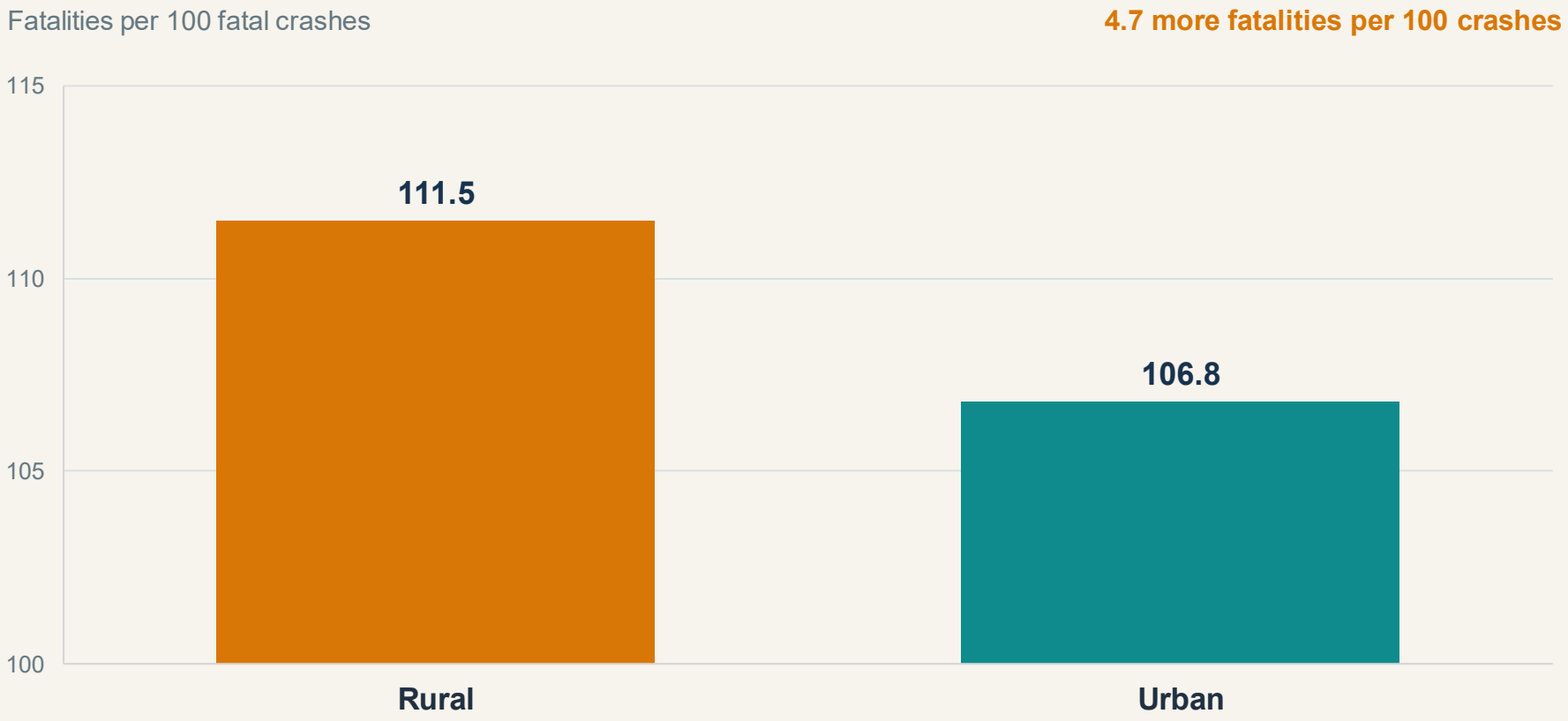
15,064

rural fatal crashes

Caption: Urban roads accounted for 22,516 fatal crashes, compared with 15,064 on rural roads.

Source: NHTSA, 2023 Fatality Analysis Reporting System.

Rural crashes involved more fatalities per crash



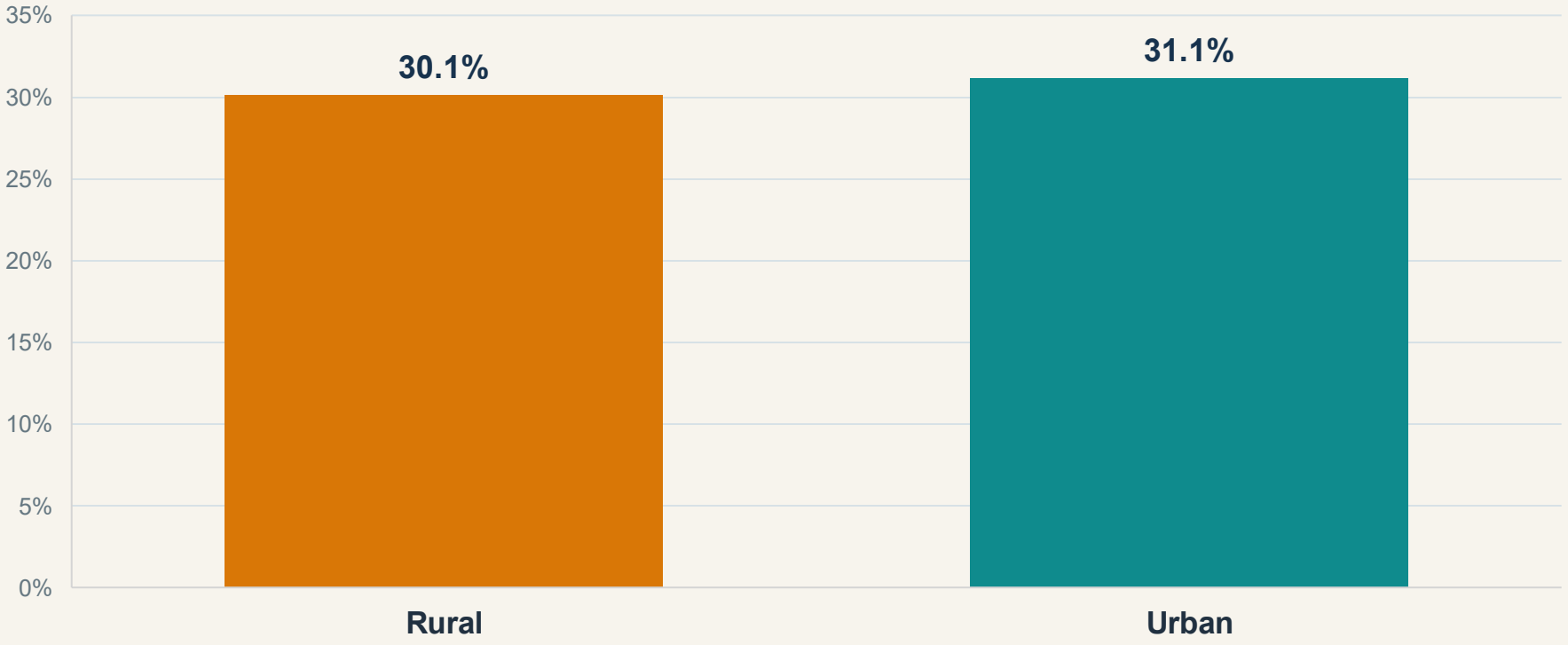
Caption: Rural crashes averaged 111.5 fatalities per 100 crashes; urban crashes averaged 106.8.

Source: NHTSA, 2023 Fatality Analysis Reporting System.

Speeding involvement is nearly equal

Share of known-status fatal crashes coded speed-related

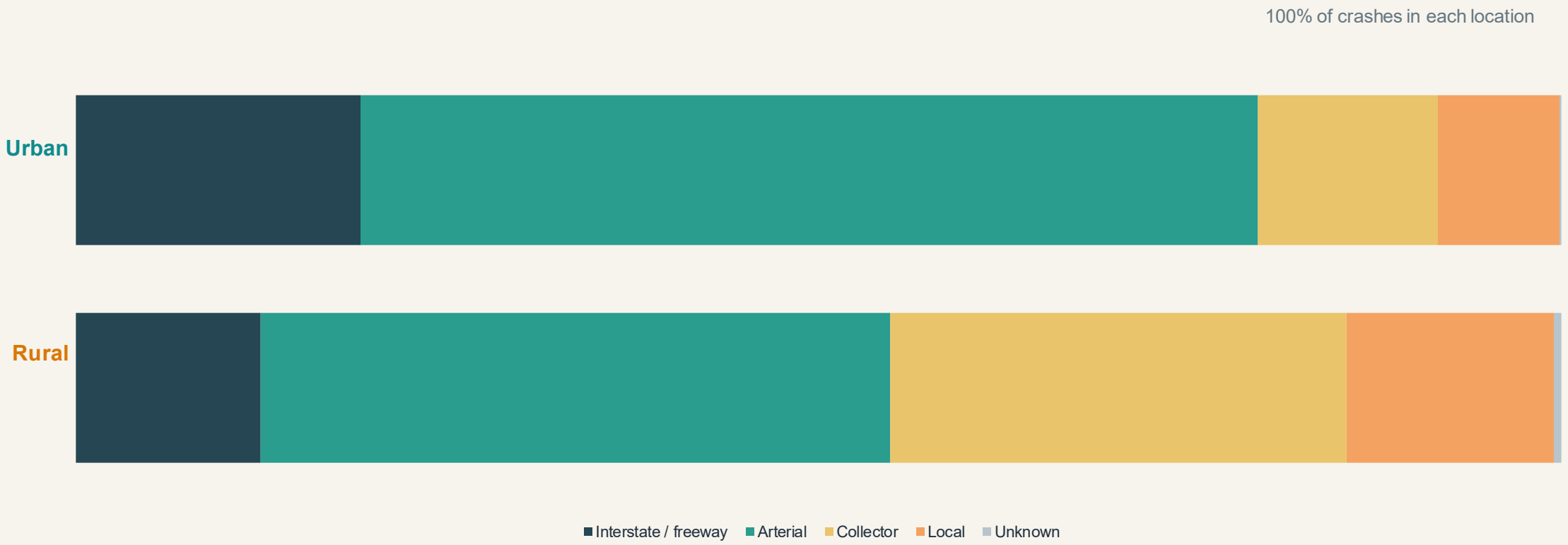
Only a 1.0 percentage-point difference



Caption: Among crashes with known speeding status, the urban rate was 31.1% and the rural rate was 30.1%.

Source: NHTSA, 2023 Fatality Analysis Reporting System.

Roadway mix differs sharply by location



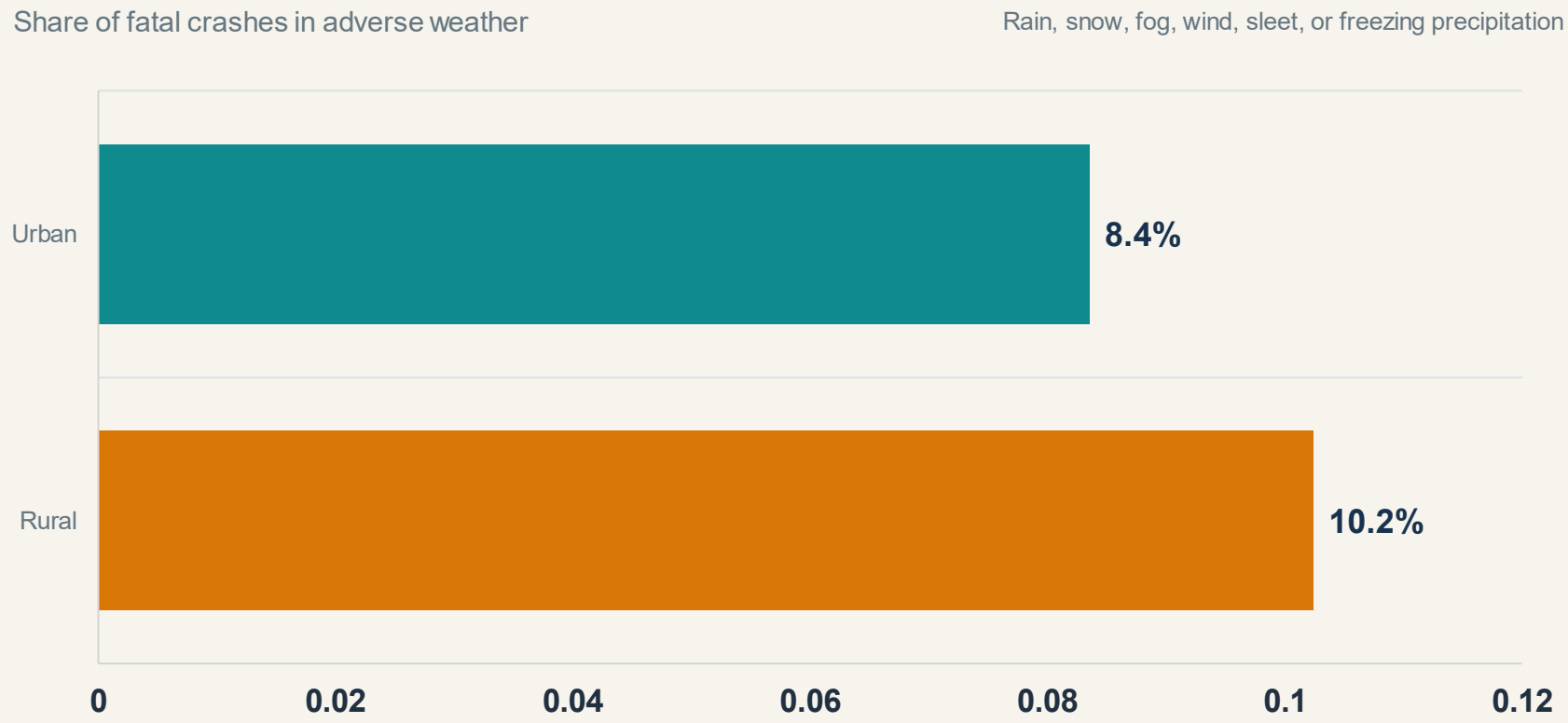
Rural: 30.7% collector roads

Urban: 60.4% arterial roads

Caption: Collectors make up 30.7% of rural fatal crashes, while arterials make up 60.4% of urban fatal crashes.

Source: NHTSA, 2023 Fatality Analysis Reporting System.

Adverse weather is more common in rural crashes



Caption: Adverse weather appears in 10.2% of rural fatal crashes and 8.4% of urban fatal crashes.

Source: NHTSA, 2023 Fatality Analysis Reporting System.

What the comparison shows

Urban roads

Contain 59.9% of classified fatal crashes

Rural roads

Have more fatalities per crash and a larger adverse-weather share

Main distinction

Roadway mix differs more than speeding involvement

Limitations

- Counts do not measure risk per mile driven
- FARS includes only crashes with a death within 30 days
- Coded variables may contain reporting error
- The analysis describes one year
- Associations do not establish causation

Sources

NHTSA FARS overview and methodology
2023 National CSV archive
FARS Analytical User's Manual
DATA 110 Project 1 README